1 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

[Total Marks: 2]

2 **(a)** $x = 4$ **[B1]**

(b) $y = 3$ **[B1]**

[Total Marks: 2]

3 **(a)** -3 **[B1]**

(b) -2 **[B1]**

[Total Marks: 2]

4 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $x = -2$ **[B1]**

[Total Marks: 3]

5 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $f(0) = 0$ **[B1]**

$x = 0$ **[B1]**

[Total Marks: 4]

6 **(a)** $x = 3$ **[B1]**

(b) $y = 1$ **[B1]**

(c) **(i)** 4 **[B1]**

(ii) 2 **[B1]**

[Total Marks: 4]

7 **(a)** **(i)** $x = 1$ **[B1]**

(ii) $y = 3$ **[B1]**

(b) $\dfrac{4}{x-1} + 3 = 0$ **[M1]**

$(-\frac{1}{3}, 0)$ **[A1]**

(c) $(0, -1)$ **[B1]**

[Total Marks: 5]

8 (a) $x = 45°$ **[B1]**

$x = 135°$ **[B1]**

(b) Draws dashed vertical lines at $x = 45°$ and $x = 135°$ **[B1]**

Correct tangent curve shape passing through $(0°, 0)$, $(90°, 0)$, and $(180°, 0)$ **[B1]**

Branches approach asymptotes cleanly without touching or curling back **[B1]**

[Total Marks: 5]

9 (a) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(b) There is a vertical asymptote at $x = 90°$ (or the curve never touches $x = 90°$) **[B1]**

(c) $x = 125°$ **[B1]**

$x = 305°$ **[B1]**

(d) 225 **[B1]**

[Total Marks: 6]

10 (a) $90°$ **[B1]**

$270°$ **[B1]**

(b) Vertical asymptote: $x = -2.5$ (or $-\frac{5}{2}$) **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(c) $4x - 1 = 0$ **[M1]**

0.25 (or $\frac{1}{4}$) **[A1]**

[Total Marks: 6]

11 (a) Vertical asymptote: $x = -2$ **[B1]**

Horizontal asymptote: $y = 1$ **[B1]**

(b) Substitute $y = 0$ to find x-intercept: $0 = \frac{3}{x+2} + 1 \implies x = -5$ **[M1]**

x-axis at $(-5, 0)$ and y-axis at $(0, 2.5)$ **[A1]**

(c) Two correct branches of the hyperbola sketched **[B1]**

Graph approaches $x = -2$ and $y = 1$ correctly without touching or crossing them **[B1]**

Graph passes through $(-5, 0)$ and $(0, 2.5)$ approximately **[B1]**

[Total Marks: 7]

12 (a) $x = -2$ **[B1]**

(b) $y = -4$ **[B1]**

(c) Sets $x = 0$ to find the y-intercept: $y = \frac{3}{0+2} - 4$ **[M1]**

y-intercept is $(0, -2.5)$ **[A1]**

Sets $y = 0$ to find the x-intercept: $0 = \frac{3}{x+2} - 4$ **[M1]**

$4 = \frac{3}{x+2} \implies 4(x+2) = 3 \implies 4x + 8 = 3$ **[M1]**

x-intercept is $(-1.25, 0)$ **[A1]**

(d) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

[Total Marks: 9]

13 (a) $b = 2$ **[B1]**

$c = -1$ **[B1]**

(b) Substitutes $(5, 0)$ into the equation: $\frac{a}{5-2} - 1 = 0$ **[M1]**

$a = 3$ **[A1]**

(c) Sets up the equation $\frac{3}{x-2} - 1 = -x + 5$ or uses GDC to find intersection of $y = f(x)$ and $y = -x + 5$ **[M1]**

$x = 3$ and $x = 5$ **[A1]**

[Total Marks: 6]

14 (a) $q = 2$ **[B1]**

$r = -1$ **[B1]**

(b) Substitutes $(3, 3)$, $q = 2$ and $r = -1$ into $f(x)$: $3 = \frac{p}{3-2} - 1$ **[M1]**

$p = 4$ **[A1]**

(c) Sets $f(x) = 0$

$\frac{4}{x-2} - 1 = 0$ [M1]

$$\frac{4}{x-2} = 1$$

x - 2 = 4[$M1$]

$(6, 0)$ **[A1]**

[Total Marks: 7]

15 (a) Two correct vertical asymptotes at $x = 90°$ and $x = 270°$ **[B1]**

Three correct branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ with correct general curvature **[B1]**

(b) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(c) $\tan x = -1.5$ **[M1]**

$x = 123.7°$ **[A1]**

$x = 303.7°$ **[A1]**

[Total Marks: 7]

16 (a) $x = -0.5$ (or $-\frac{1}{2}$) **[B1]**

$x = 3$ **[B1]**

(b) $x = 2$ **[B1]**

(c) As x

$\pm\infty$, $h(x)$ increases or decreases without bound (does not approach a constant value) [B1]

Mentions that the degree of the numerator (2) is greater than the degree of the denominator (1), or that the curve approaches an oblique/sloping line $y = 2x - 1$ **[B1]**

(d) Finds local maximum at $(0.27, 4.07)$ and local minimum at $(3.73, 1.93)$ using GDC **[M1]**

$1.93 < k < 4.07$ **[A1]**

[Total Marks: 7]

17 (a) (i) $x = 1.5$ **[B1]**

(ii) $y = 2$ **[B1]**

(b) $y = 2$ is a horizontal asymptote so the curve never reaches $y = 2$ **[B1]**

(c) Vertical asymptote: $x = -1.5$ **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(d) $\frac{5}{x - 1.5} + 2 = 3.4$ or $\frac{5}{x - 1.5} = 1.4$ **[M1]**

$x - 1.5 = \frac{5}{1.4}$ **[M1]**

5.07 **[A1]**

[Total Marks: 8]

18 **(a)** $q = 5$ **[B1]**

$r = -3$ **[B1]**

(b) Substitutes $x = 7, y = -2$ and their values of q and r into the equation: $-2 = \dfrac{p}{7-5} - 3$ **[M1]**

$-2 = \dfrac{p}{2} - 3 \implies \dfrac{p}{2} = 1$ **[M1]**

$p = 2$ **[A1]**

(c) Sets $y = \dfrac{2}{x-5} - 3$ and attempts to make x the subject, or swaps x and y **[M1]**

$y + 3 = \dfrac{2}{x-5} \implies x - 5 = \dfrac{2}{y+3}$ **[M1]**

$x = \dfrac{2}{y+3} + 5$ **[M1]**

$f^{-1}(x) = \dfrac{2}{x+3} + 5$ or equivalent **[A1]**

(d) $x = -3$ **[B1]**

$y = 5$ **[B1]**

(Strict follow through from their answers in part (a): $x = r, y = q$)

[Total Marks: 11]